

lecture_05

August 1, 2026

1 Lecture 5: Spreadsheet Solver Method

Learning Objectives

By the end of this lecture, you will be able to: 1. Set up a linear program in Microsoft Excel using a structured spreadsheet layout 2. Configure the Solver add-in (objective, decision variables, constraints) and interpret its output

Prerequisites: LP formulation (Lecture 3), Graphical Method (Lecture 4)

Estimated time: 90 minutes (including in-class exercises)

Note: This lecture uses **Microsoft Excel with the Solver Add-in**. If Solver is not visible under the *Data* tab, go to *File* → *Options* → *Add-ins* → *Manage Excel Add-ins* → *check Solver Add-in* → *OK*.

1.1 Why Use a Spreadsheet Solver?

In Lecture 4 we solved LPs by hand — the graphical method for 2-variable problems. Real transportation problems, however, involve tens or hundreds of variables and constraints. Manually pivoting through a tableau becomes impractical.

The **Excel Solver** is a practical, widely-used tool that:

- Requires no programming — ideal for quick, one-off analyses
- Is used extensively in industry: logistics planners, fleet managers, and traffic engineers regularly use it for day-to-day decisions

In this lecture we will solve two problems in class and you will practise three problems as take-away exercises.

1.2 The Excel Solver Workflow

Setting up any LP in Excel follows the following template:

Section	What you do in Excel
1. Data	Enter all problem parameters (costs, capacities, demands) in clearly labelled cells

Section	What you do in Excel
2. Decision Variables	Reserve a row/column of cells for $x_1, x_2, \dots$ — start with value 0
3. Objective Function	Write the objective function formula: =SUMPRODUCT(coefficients, decision_vars)
4. Constraint cells	Write LHS formulas for each constraint; enter RHS values separately
5. Solver Box	Set objective → set variable cells → add constraints → choose <i>Simplex LP</i> → Solve

1.2.1 Solver Dialog Settings

- **Step 1** - Set the *Objective* box to the cell containing your objective formula
- **Step 2** - Set the *To* option to Min or Max
- **Step 3** - Set the *By Changing Variable Cells* to the range of cells holding decision variables
- **Step 4** - Set the *Subject to the Constraints* by clicking *Add* for each constraint
- **Step 5** - Set the *Solving Method*; choose **Simplex LP** for linear programs
- **Step 6** - After solving select *Keep Solver Solution* (assuming the solver has found the solution)

1.3 In-Class Exercises

1.3.1 Exercise 1: Freight Mode Allocation

Problem Statement A logistics company operates a Chennai–Delhi freight corridor. It can ship cargo either by truck or by rail. The planning horizon is one week.

	Truck	Rail
Revenue (/tonne)	4,500	3,200
Operating cost (/tonne)	2,800	1,400
CO emissions (kg/tonne)	62	18

Constraints:

1. Total capacity: at most 500 tonnes can be shipped in a week (combined)
2. Rail capacity: at most 300 tonnes by rail (track allocation limit)
3. Minimum service: at least 100 tonnes must go by truck (contractual obligation)
4. Emission cap: total CO must not exceed 24,000 kg
5. Non-negativity: tonnes shipped ≥ 0

Maximize total profit (revenue – operating cost).

Step 1 — Formulate the LP Let: - x_T = tonnes shipped by truck - x_R = tonnes shipped by rail

Objective:

$$\max Z = (4500 - 2800) x_T + (3200 - 1400) x_R = 1700 x_T + 1800 x_R$$

Subject to:

$$\begin{aligned} x_T + x_R &\leq 500 && \text{(Capacity)} \\ x_R &\leq 300 && \text{(Rail cap)} \\ x_T &\geq 100 && \text{(Min truck)} \\ 62 x_T + 18 x_R &\leq 24000 && \text{(Emission cap)} \\ x_T, x_R &\geq 0 \end{aligned}$$

Step 2 — Excel Spreadsheet Layout Set up your spreadsheet exactly as follows:

	A	B	C	
1	FREIGHT MODE ALLOCATION MODEL			
2				
3	Parameter	Truck	Rail	
4	Profit (/tonne)	1700	1800	
5	CO2 (kg/tonne)	62	18	
6				
7	Decision Variables	Truck	Rail	
8	Tonnes shipped	0	0	← Solver changes these
9				
10	Objective			
11	Total Profit ()	=SUMPRODUCT(B4:C4, B8:C8)		
12				
13	Constraints	LHS	RHS	Slack
14	Total capacity	=SUM(B8:C8)	500	=C14-B14
15	Rail capacity	=C8	300	=C15-B15
16	Min truck	=B8	100	=B16-C16
17	Emission cap	=SUMPRODUCT(B5:C5,B8:C8) 24000 =C17-B17		

Step 3 — Solver Configuration

- **Set Objective:** B11 → Max
 - **Variable Cells:** B8:C8
 - **Constraints:**
 - B14 C14
 - B15 C15
 - B16 C16
 - B17 C17
 - B8:C8 0
 - **Method:** Simplex LP → Solve
-

Step 4 — Expected Results & Interpretation After solving, verify:

Variable	Value
x_T (truck)	200 tonnes
x_R (rail)	300 tonnes
Total Profit	8,80,000

Managerial insight: Both the total capacity constraint and the rail capacity constraint are binding at optimum — every available tonne of capacity is being used, and the rail allocation is fully exhausted. The minimum truck obligation and emission cap are both slack: the operator is already exceeding the minimum truck requirement, and emissions are well within the cap, so neither constraint is a binding concern at current demand levels.

1.3.2 Exercise 2

Problem Statement A signalised intersection in Kolkata has 4 phases. The traffic engineer wants to allocate green time (in seconds) across phases to maximize total vehicle throughput (vehicles/cycle).

Phase	Movement	Saturation flow (veh/s)	Min green (s)	Max green (s)
1	Northbound through	0.52	10	45
2	Southbound through	0.48	10	45
3	Eastbound through	0.35	8	35
4	Westbound through	0.30	8	35

Constraints:

1. Cycle length = 90 seconds (fixed): $g_1 + g_2 + g_3 + g_4 = 90$

(we subtract a fixed lost time of 2s/phase $\rightarrow$ effective green budget = $90 - 4 \times 2 = 82s$)

2. Minimum green: $g_i \geq g_i^{\min}$ for each phase i
3. Maximum green: $g_i \leq g_i^{\max}$ for each phase i

Maximize total throughput = $\sum_i s_i \cdot g_i$ where s_i is saturation flow.

Step 1 — LP Formulation Let g_i = effective green time (seconds) allocated to phase i .

Objective:

$$\max Z = 0.52 g_1 + 0.48 g_2 + 0.35 g_3 + 0.30 g_4$$

Subject to:

$$\begin{aligned} g_1 + g_2 + g_3 + g_4 &= 82 && \text{(Cycle budget)} \\ g_i &\geq g_i^{\min}, \quad i = 1, 2, 3, 4 && \text{(Min green)} \\ g_i &\leq g_i^{\max}, \quad i = 1, 2, 3, 4 && \text{(Max green)} \end{aligned}$$

Note: This is an LP with one **equality** constraint. In Solver, enter it as LHS = RHS (use the = option in the Add Constraint dialog).

Step 2 — Excel Spreadsheet Layout

	A	B	C	D	E
1	SIGNAL TIMING OPTIMISATION				
2					
3	Parameter	Ph 1	Ph 2	Ph 3	Ph 4
4	Sat. flow (veh/s)	0.52	0.48	0.35	0.30
5	Min green (s)	10	10	8	8
6	Max green (s)	45	45	35	35
7					
8	Decision Vars	Ph 1	Ph 2	Ph 3	Ph 4
9	Green time (s)	0	0	0	0 ← Solver changes
10					
11	Objective				
12	Throughput (veh)	=SUMPRODUCT(B4:E4, B9:E9)			
13					
14	Constraints	LHS	RHS		
15	Cycle budget	=SUM(B9:E9)	82	← equality	
16	Min Ph1	=B9	10		
17	Min Ph2	=C9	10		
18	Min Ph3	=D9	8		
19	Min Ph4	=E9	8		
20	Max Ph1	=B9	45		
21	Max Ph2	=C9	45		
22	Max Ph3	=D9	35		
23	Max Ph4	=E9	35		

Step 3 — Solver Configuration

- **Set Objective:** B12 → Max
 - **Variable Cells:** B9:E9
 - **Constraints:** B15 = C15 (equality!), then min/max bounds
 - **Method:** Simplex LP → Solve
-

Step 4 — Expected Results & Interpretation

Phase	Optimal green (s)
g_1	45 (at upper bound)
g_2	21
g_3	8 (at lower bound)
g_4	8 (at lower bound)
Throughput	38.7 veh/cycle

Interpretation: The solver assigns maximum green to Phase 1 (highest saturation flow) and Phase 2 gets the remaining budget after minimums are satisfied for Phases 3 and 4.

1.4 Take-Away Exercises

Work through the following steps: 1. Write the objective function 2. Write all constraints with correct signs 3. Build the spreadsheet layout following the 5-step template 4. Run Solver with Simplex LP 5. Record optimal outcomes 6. Make managerial inferences

1.4.1 Exercise 1

The Chennai Metropolitan Transport Corporation (MTC) operates two routes: - Route A (high-demand urban corridor): requires at least 3,200 passenger-km/day - Route B (suburban feeder): requires at least 1,800 passenger-km/day

MTC can deploy two bus types:

	Standard Bus	Articulated Bus
Capacity (pass-km/day per bus)	400	750
Operating cost (/day per bus)	8,500	14,000
Maintenance cost (/day per bus)	1,200	2,500
Driver requirement (drivers/bus)	1	2

MTC must satisfy five operational constraints. On the demand side, Route A must deliver at least 3,200 pass-km/day and Route B at least 1,800 pass-km/day. On the supply side, the daily operating budget cannot exceed 2,00,000, the driver pool is limited to 60, and the total fleet to 25 buses.

Assume each bus type can be split between routes. Let: - x_1 = standard buses deployed (total across both routes) - x_2 = articulated buses deployed (total across both routes)

Minimize total daily cost.

1.4.2 Exercise 2

A state transport depot in Delhi maintains a fleet of 200 buses and needs to blend two grades of diesel to meet fleet specifications:

Specification	Fleet Requirement
Cetane number	48
Sulphur content	15 ppm
Density	Between 820 and 845 kg/m ³

Two diesel grades are available:

Property	Grade A (High-quality)	Grade B (Standard)
Cetane number	55	42
Sulphur (ppm)	8	22
Density (kg/m ³)	830	850
Cost (/litre)	92	79

The depot needs 10,000 litres/day total. Blended properties are volume-weighted averages.

Let: - x_A = litres of Grade A per day - x_B = litres of Grade B per day

Minimize daily fuel cost.

1.4.3 Exercise 3

The Kerala State Highway Authority has a budget of 90 crore to widen single-lane stretches of state highway to two lanes. There are 5 candidate stretches identified by the annual road-safety audit. Widening a stretch reduces its annual accident count, based on engineering studies of crash rates on single- vs. two-lane sections of comparable traffic volume.

Stretch	Widening Cost (Cr)	Annual Accidents Prevented	Length (km)
S1 — Kochi– Aluva	32	18	14
S2 — Thrissur– Guruvayur	24	11	9
S3 — Kollam– Punalur	28	14	12
S4 — Kottayam– Erattupetta	18	8	7

Stretch	Widening Cost (Cr)	Annual Accidents Prevented	Length (km)
S5 — Kannur– Thalassery	22	12	10

Maximize total accidents prevented per year by deciding what fraction $y_s \in [0, 1]$ of each stretch to widen (a partially-funded stretch widens a proportional length, preventing a proportional share of accidents — a standard assumption in multi-year phased capital planning).

1.5 Further Reading

- Transportation Research Board (2010). *Highway Capacity Manual* — signal timing optimization background
- Winston, W.L. (2004). *Operations Research: Applications and Algorithms* — Excel Solver examples in Chapters 3–4
- For Solver documentation: *File* → *Help* within Excel, or support.microsoft.com/solver